Gaussian Process + Information Theory Review

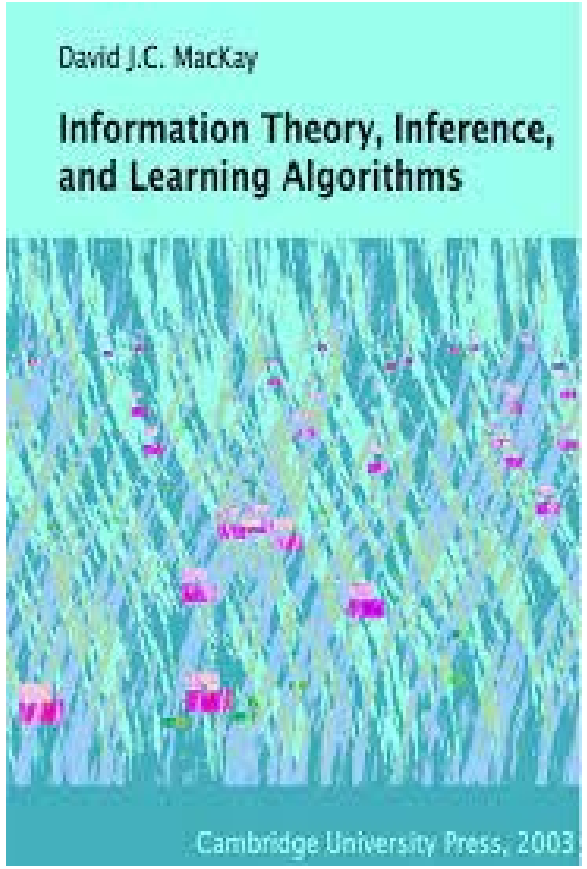
Lecture 4

Friday March 13, 2026

NEXT: **Lecture 5**, Tuesday, March 17: 16:30–18:00 CET
Prof. Albert (in-person)

Suggested Reference: It's free!

https://www.inference.org.uk/itprnn/book.pdf



Bayesian Inference + Information Theory

1 Introduction to Information Theory
2 Probability, Entropy, and Inference
3 More about Inference

I Data Compression
4 The Source Coding Theorem
5 Symbol Codes
6 Stream Codes
7 Codes for Integers

II Noisy-Channel Coding
8 Dependent Random Variables
9 Communication over a Noisy Channel
10 The Noisy-Channel Coding Theorem
11 Error-Correcting Codes and Real Channels

III Further Topics in Information Theory
12 Hash Codes
13 Binary Codes
14 Very Good Linear Codes Exist
15 Further Exercises on Information Theory
16 Message Passing
17 Constrained Noiseless Channels
18 Crosswords and Codebreaking
19 Why have Sex?

IV Probabilities and Inference
20 An Example Inference Task: Clustering
21 Exact Inference by Complete Enumeration
22 Maximum Likelihood and Clustering
23 Useful Probability Distributions
24 Exact Marginalization
25 Exact Marginalization in Trellises
26 Exact Marginalization in Graphs
27 Laplace's Method
28 Model Comparison and Occam's Razor
29 Monte Carlo Methods
30 Efficient Monte Carlo Methods
31 Ising Models
32 Exact Monte Carlo Sampling
33 Variational Methods
34 Independent Component Analysis
35 Random Inference Topics
36 Decision Theory
37 Bayesian Inference and Sampling Theory

V Neural networks
38 Introduction to Neural Networks
39 The Single Neuron as a Classifier
40 Capacity of a Single Neuron
41 Learning as Inference
42 Hopfield Networks
43 Boltzmann Machines
44 Supervised Learning in Multilayer Networks
45 Gaussian Processes
46 Deconvolution

Gaussian Process Regression

A Local Method for Machine Learning

Gaussian Process as Path Integral

Most valuable integral in physics,

$$\int \exp\left(-\frac{1}{2}x \cdot \underbrace{A} \cdot x + J \cdot x\right) d^n x = \sqrt{\frac{(2\pi)^n}{\det A}} \exp\left(\frac{1}{2} J \cdot \underbrace{A^{-1}} \cdot J\right)$$

We need to talk about this object

Here A is a real positive definite symmetric matrix.

This integral is performed by diagonalization of A with an orthogonal transformation

$$D = O^{-1} A O = O^T A O$$

where D is a diagonal matrix and O is an orthogonal matrix. This decouples the variables and allows the integration to be performed as n one-dimensional integrations.

$$\int \exp\left[\int d^4x \left(-\frac{1}{2}\varphi \hat{A} \varphi + J\varphi\right)\right] D\varphi \;\propto\; \exp\left(\frac{1}{2}\int d^4x\, d^4y J(x) \overbrace{D(x-y)} J(y)\right)$$

https://en.wikipedia.org/wiki/Common_integrals_in_quantum_field_theory

Comparing Kernels and Matrices

Hermitian (symmetric): $k^*(x', x) = k(x, x')$ $\left[\mathbf{A}^T = \mathbf{A}\right]$

Positive-definite: $\int_{\mathcal{D}} dx \int_{\mathcal{D}} dx' \, f^*(x) k(x, x') g(x') \geq 0, \forall f, g \in \mathcal{H}$ $\left[\mathbf{x}^T \mathbf{A} \mathbf{y} \geq 0\right]$

Eigenfunctions of the kernel: $\int_{\mathcal{D}} dx' \; k(x, x') \cdot \phi_i(x') = \lambda_i \phi_i(x)$ $\left[\mathbf{A}\mathbf{v}_i = \lambda_i \mathbf{v}_i\right]$

Mercer representation of the kernel: $k(x, x') = \sum_{i=1}^{\infty} \lambda_i \phi_i(x) \phi_i^*(x')$ $\left[\mathbf{A} = \sum_{i=1}^{N} \lambda_i \mathbf{v}_i \mathbf{v}_i^T\right]$

Kernel Methods in Machine Learning

If the data space is NOT a (Euclidean) vector space, then find a *non-linear* mapping to a vector space (Reproducing Kernel Hilbert Space).

Single data point Encoder: $\vec{w} = \Phi(\vec{x})$ **Not used explicitly**!

Inner product on feature space: $\kappa(\vec{x}, \vec{x}') = \left\langle \Phi(\vec{x}), \Phi(\vec{x}') \right\rangle = w^i g_{ij} w^j$

Designing or learning this similarity measure is the key task.

Form Gram matrix: $G = A^T A \quad \Rightarrow \quad G_{ij} = \left\langle \Phi(\vec{x}_i), \Phi(\vec{x}_j) \right\rangle = \kappa(\vec{x}_i, \vec{x}_j)$

Encodes all information in data.

Reproducing property of kernel: $f(\vec{x}) = \left\langle \vec{x} \mid f \right\rangle = \sum_{m=1}^{M} \alpha_m \kappa(\vec{x}_m, \vec{x})$

Can form all functions for regression and classification.

Gaussian Process: Nomenclature

Input space: $\mathcal{M}$
Sampling grid: $\mathbf{x} \in \mathcal{M}$ (the 'design')
Query point(s): x_*, $\mathbf{x}_* \in \mathcal{M}$

Output space: $\mathcal{N}$
Generative function: $f \in C^2(\mathcal{N})$ (unknown)
Measurements of f: $\mathbf{y} \in \mathcal{N}$

Interpolated function: $\mathbf{f}_* = f(\mathbf{x}_*)$

This is the posterior mean function from the GPR.

Prior functional variance: σ_f^2

Spatial bandwidth of prior: τ

Measurement variance: σ_0^2

ROI weight function: $w(x)$

The Gaussian Process on the function space, $\mathcal{H}$, is defined by placing a *Gaussian measure* on the output space, $\mathcal{N}$, at each point of the input space, $\mathcal{M}$.

In general, $\mathcal{M} \xrightarrow{f} \mathcal{N}$

Scalar function, $\mathcal{M} \xrightarrow{f} \mathbb{R}$

Inner product in the feature space, $\mathcal{H}$, *pulled back* to two points in the input space:

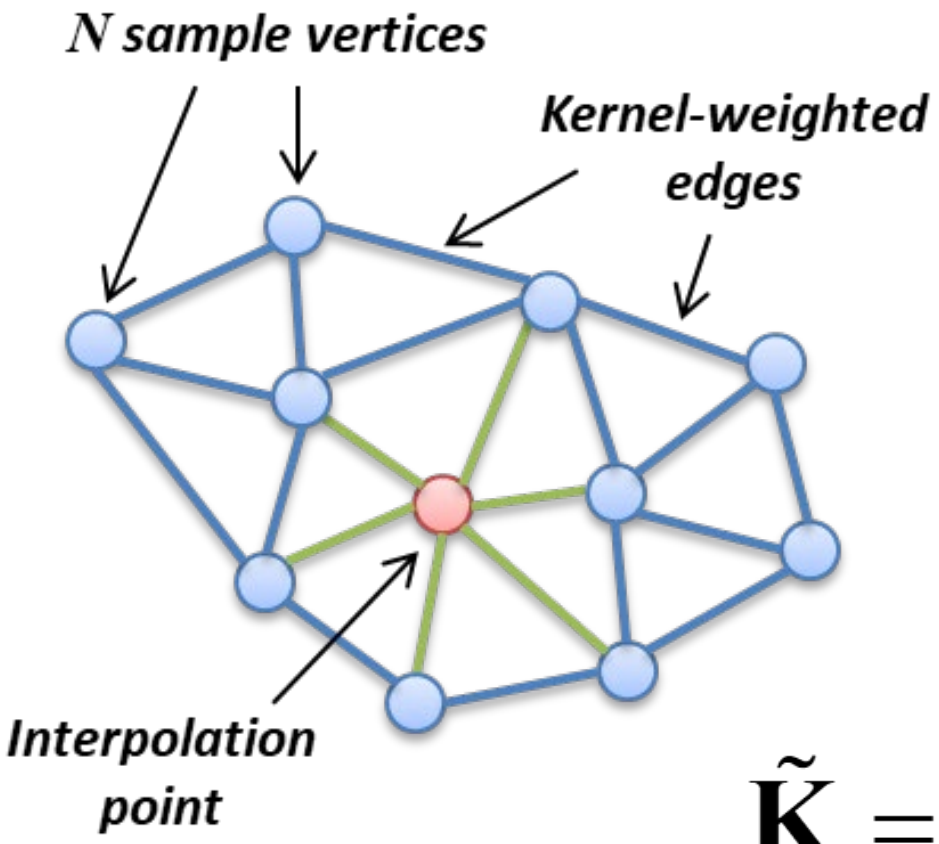
$$k : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$$

Place a probability measure on these inner products

$$f = \sum_{i=1}^{\infty} \boxed{\langle f, \phi_i \rangle} \phi_i$$

The function, f, belongs to a Hilbert space, $\mathcal{H}$, *not* the space of the measurements, $\mathcal{N}$.

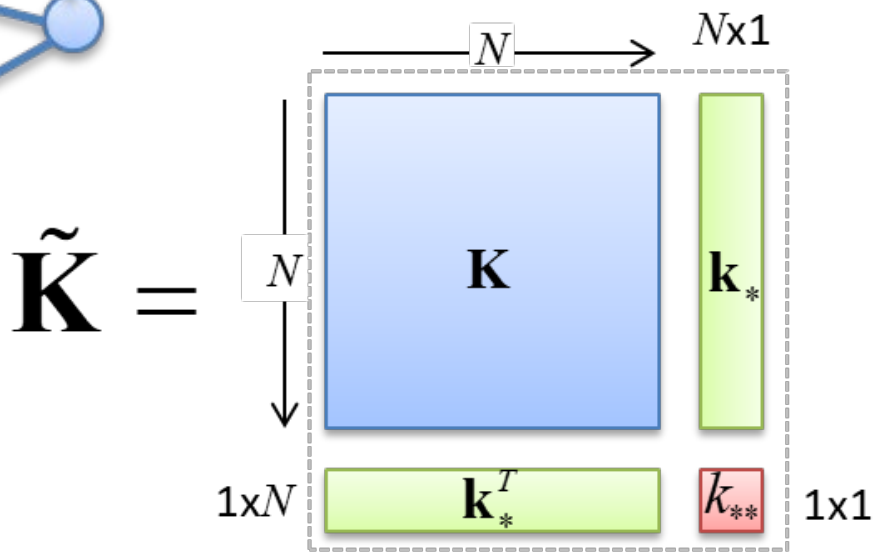
Gaussian Process Regression

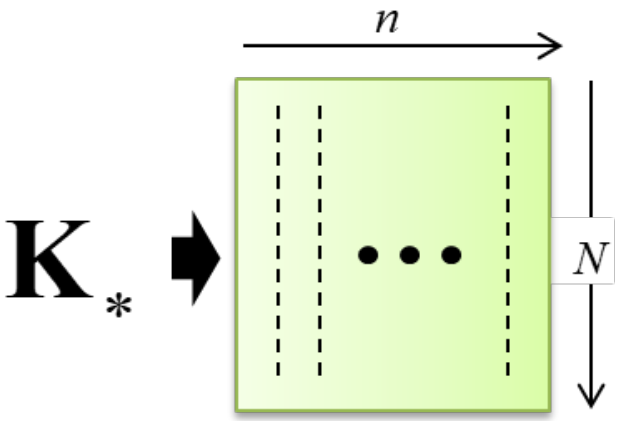


Example,

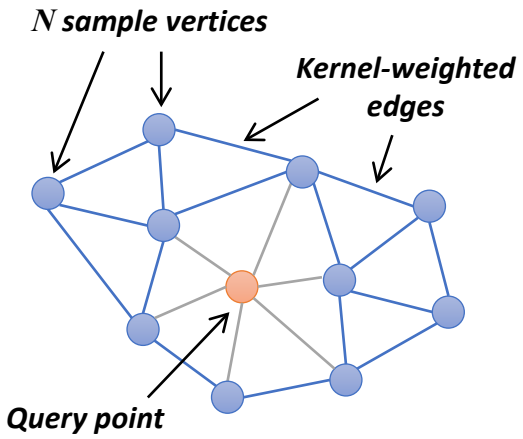
Kernel: $K(\mathbf{x}, \mathbf{x}' \mid \tau) = \exp \frac{\left|\mathbf{x} - \mathbf{x}'\right|^2}{2\tau}$

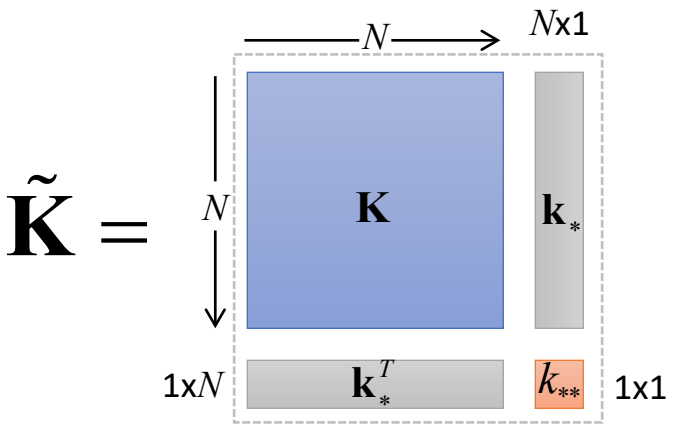
Radial Basis Function (~Diffusion Green's function)

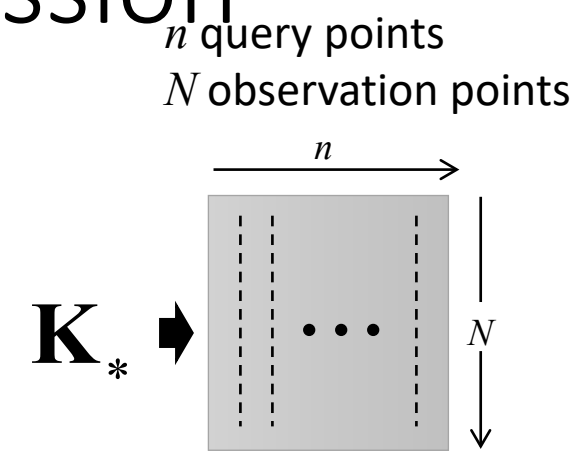




Gaussian Process Regression







Weight perspective:

GPR equations,

$$p(f_* \mid x_*; \mathbf{y}, \mathbf{x}) = \mathcal{N}(f_* \mid \mu_*, \sigma_*^2)$$

$$\mu_* = \mathbf{k}_*^T \left(\mathbf{K} + \sigma_0^2 \mathbf{I}_N\right)^{-1} \mathbf{y}$$

$$\sigma_*^2 = \sigma_f^2 - \mathbf{k}_*^T \left(\mathbf{K} + \sigma_0^2 \mathbf{I}_N\right)^{-1} \mathbf{k}_*$$

$$\mu_* = \mathbf{w}_*^T \mathbf{y},$$

where $\mathbf{w}_* = \left(\mathbf{K} + \sigma_0^2 \mathbf{I}_N\right)^{-1} \mathbf{k}_*$

$$\sigma_*^2 = k_{**} - \mathbf{w}_*^T \left(\mathbf{K} + \sigma_0^2 \mathbf{I}_N\right) \mathbf{w}_*$$

Interpretation: Estimate the function, $\hat{f}$, at the query point, x_*, using a linear combination of the observations, $\mathbf{y}(\mathbf{x})$, to form a weighted local average:

$$\hat{f}(x_*) = \sum_{i=1}^{N} w_i(x_*) \cdot y(x_i)$$

$$w_i(x_*) = \left[\mathbf{k}_*^T \left(\mathbf{K} + \sigma_0^2 \mathbf{I}_N\right)^{-1}\right]_i$$

Normalization: $\sum_{i=1}^{N} w_i(x_*) = 1$

Gaussian Process Regression

The *marginalization property* of the GP yields the joint probability over a finite set of measurements and interpolation points:

$$\begin{pmatrix} \mathbf{y} \\ f_* \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma} = \begin{pmatrix} \mathbf{K} + \sigma_n^2 \mathbf{I}_N & \mathbf{k}_* \\ \mathbf{k}_*^T & k_{**} + \sigma_n^2 \end{pmatrix} \right)$$

$$\boldsymbol{\Sigma} = \tilde{\mathbf{K}} + \underline{\sigma_0^2} \mathbf{I}_{N+1}$$

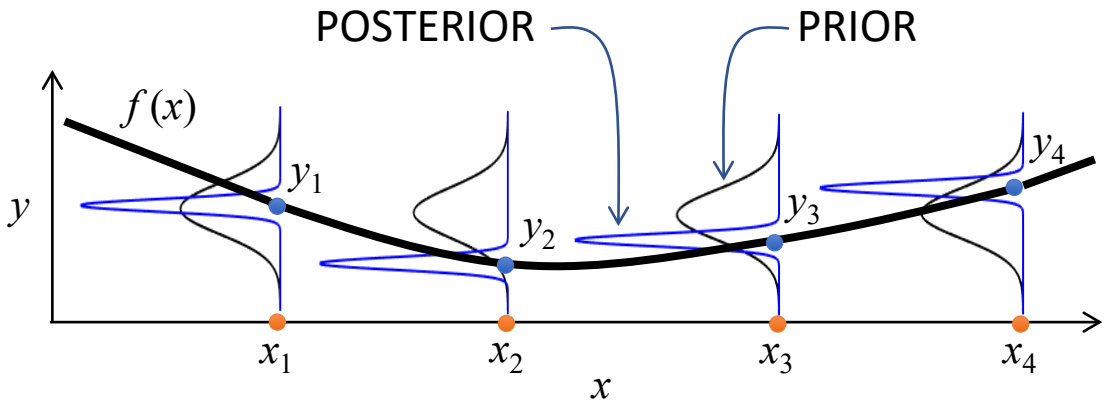
Uncorrelated measurement noise (diagonal only)

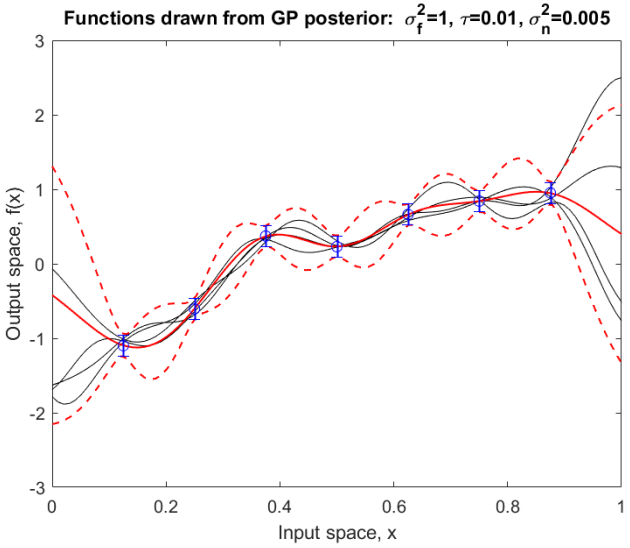
Using the inverse partitioned matrix identities, rewrite the joint probability as a conditional probability:

$$p(f_* \mid x_*; \mathbf{y}, \mathbf{x}) = \mathcal{N}(f_* \mid \mu_*, \sigma_*^2)$$

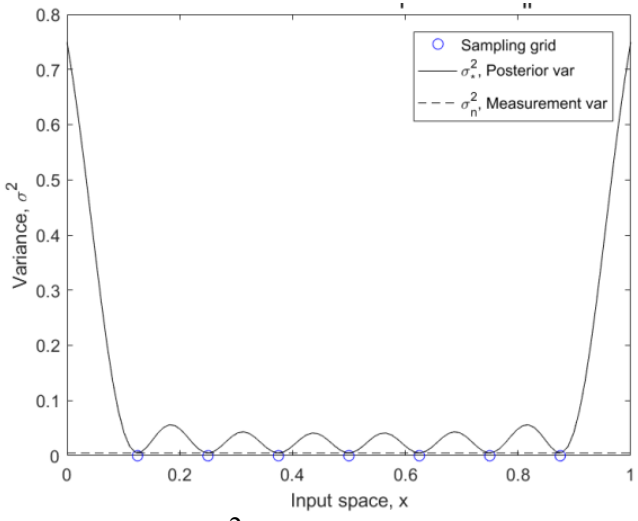
$$\mu_* = \mathbf{k}_*^T \left(\mathbf{K} + \sigma_0^2 \mathbf{I}_N \right)^{-1} \mathbf{y}$$

$$\sigma_*^2 = \sigma_f^2 - \mathbf{k}_*^T \left(\mathbf{K} + \sigma_0^2 \mathbf{I}_N \right)^{-1} \mathbf{k}_*$$



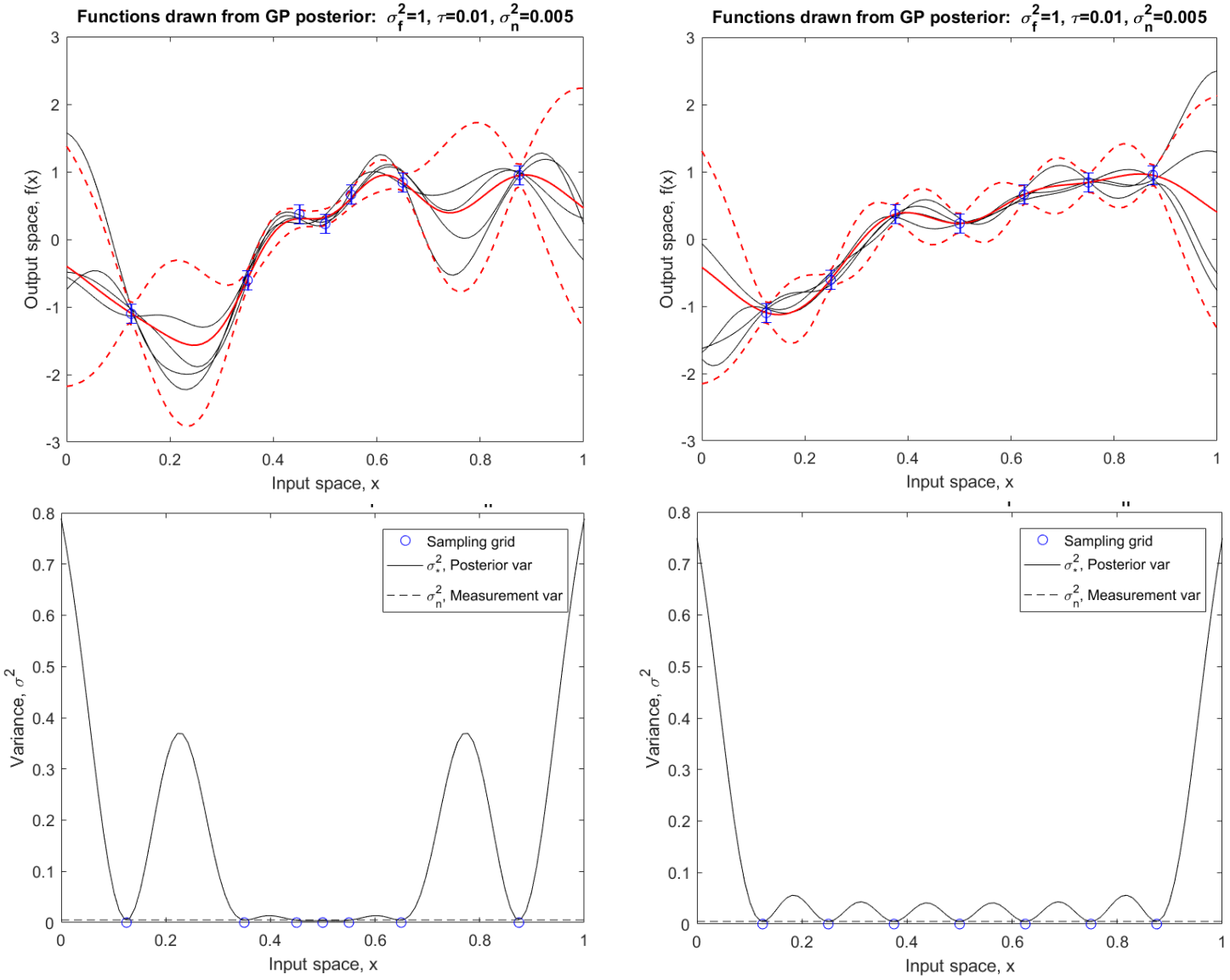


Posterior variance, σ^2



Kernel: $K(x, x' \mid \tau) = \exp \dfrac{|x - x'|^2}{2\tau}$

Gaussian Process Regression



GPR on a Sphere: Diffusion kernel

Find a kernel for the Gaussian Process that performs a meaningful statistical operation (*e.g.*, local averaging using geodesic distance) on the input space by using diffusion on a manifold:

Input space

$\mathcal{M}$

$$\left(\partial_\tau - \Delta_g\right) G(x,x';\tau) = \delta(x-x')$$

$$\text{IC}: G(x,x';\tau=0) = \delta(x-x')$$

g, metric tensor on the input space, $\mathcal{M}$.

Δ_g, Laplace-Beltrami operator on $\mathcal{M}$.

Isotropic, homogeneous case:

$$k\left(x,x' \mid \sigma_f^2,\tau\right) = \sigma_f^2 \cdot \frac{G\left(x-x';\tau\right)}{G\left(0;\tau\right)}$$

General case:

$$k\left(x,x' \mid \sigma_f^2,\tau\right) = \sigma_f^2 \cdot \frac{G\left(x,x';\tau\right)}{\sqrt{G\left(x,x;\tau\right)\cdot G\left(x',x';\tau\right)}}$$

R & W call this "vertical rescaling" - Eq.4.35, p.95, GPML.

Laplace-Beltrami operator:

$$\begin{aligned}\Delta_g f &= \operatorname{div}(\nabla f)\\ &= \frac{1}{\sqrt{|g|}}\partial_i\left(\sqrt{|g|}g^{ij}\partial_j f\right)\\ &= g^{ij}\left(\partial_i\partial_j - \Gamma_{ij}^k\partial_k\right) f\end{aligned}$$

For the 2-sphere:

$$x = (\theta,\phi)$$
$$x' = (\theta',\phi')$$

$$\Delta_{S^2} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

$$\Delta_{S^2} P_\ell(\cos\gamma_{ij}) = \ell(\ell+1)P_\ell(\cos\gamma_{ij})$$

$$G_\tau\left(\theta,\phi;\theta',\phi'\right) = \frac{1}{4\pi}\sum_{l=0}^{\infty}(2l+1)\cdot e^{-l(l+1)\tau}\cdot P_l(\cos\gamma)$$

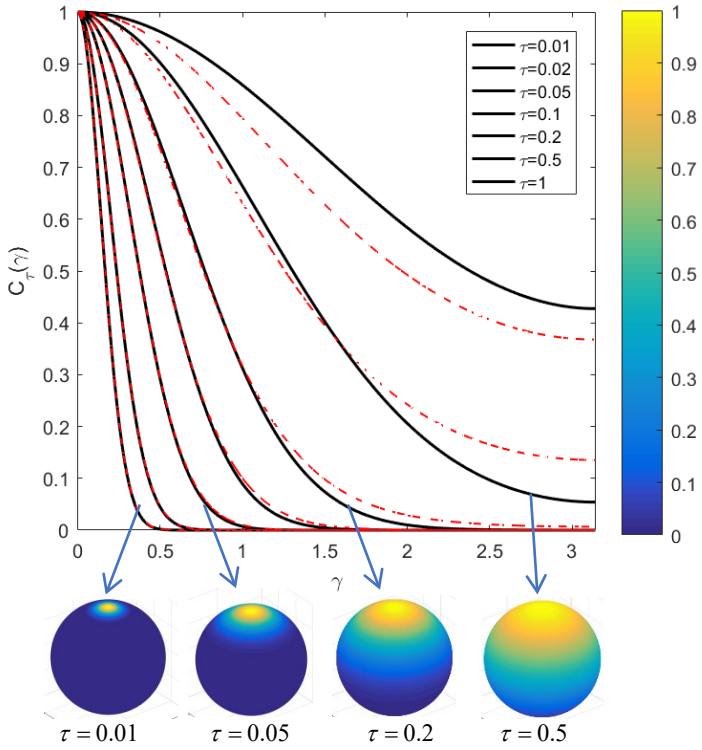
$$\cos\gamma = \cos(\phi-\phi')\cdot\sin\theta\sin\theta' + \cos\theta\cos\theta'$$

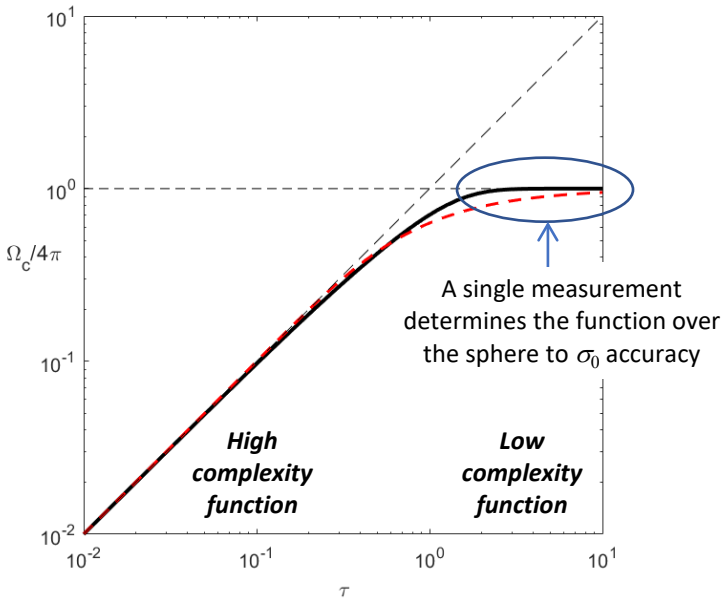
Let's call this the "Legendre kernel" (*Spherical kernel* is already taken.)

$$\boxed{k\left(\theta,\phi;\theta',\phi' \mid \sigma_f^2,\tau\right) = \sigma_f^2 \cdot \frac{\sum_{l=0}^{\infty}(2l+1)\cdot e^{-l(l+1)\tau}\cdot P_l(\cos\gamma)}{\sum_{l=0}^{\infty}(2l+1)\cdot e^{-l(l+1)\tau}}}$$

NOTE: An intuitive understanding of this kernel is that the *Radial Basis Function* (RBF) or squared exponential kernel has been "bent" on to the surface of the sphere. The RBF kernel constitutes local averaging in a flat or Euclidean space. *The Legendre kernel* implements local averaging w.r.t. the geodesics on the sphere.

Green's function or kernel on sphere

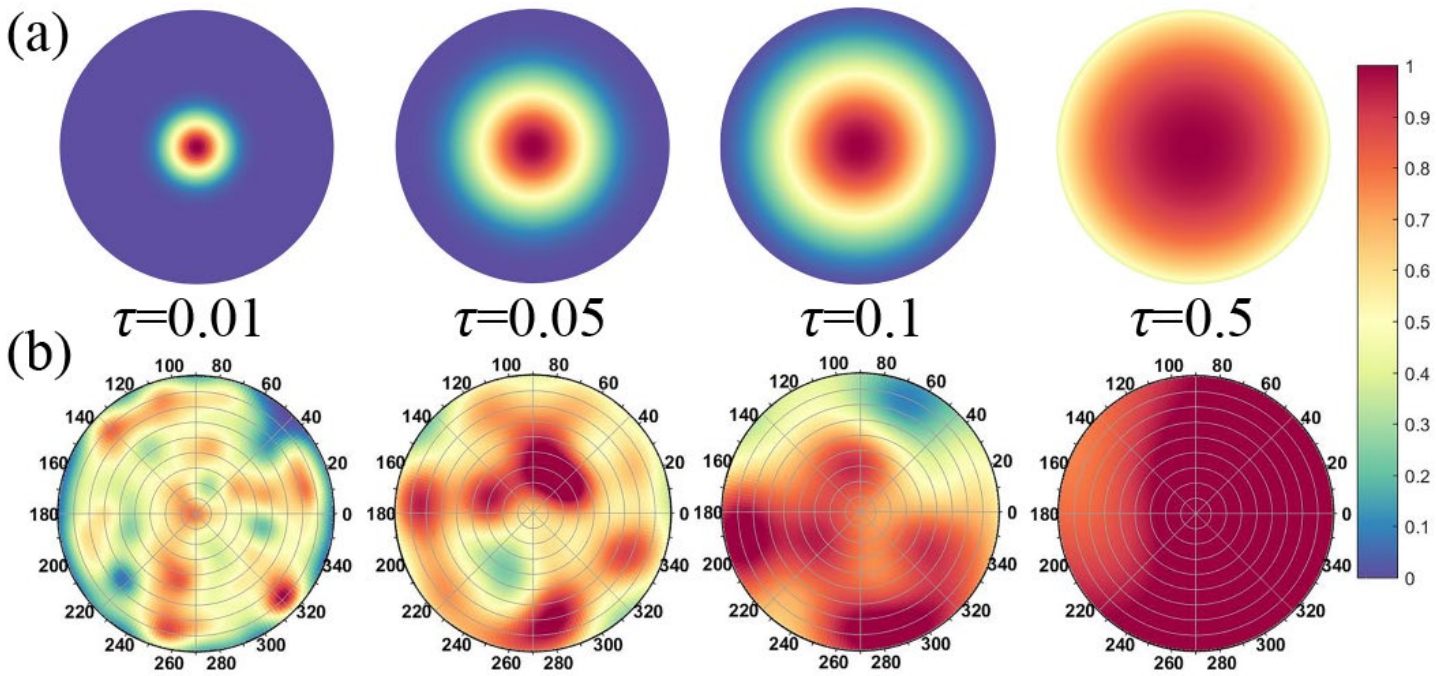




Gaussian Process on the Sphere

Random functions on a sphere:

$$\sigma_n^2 = 0.01;$$



Gaussian Process Regression (GPR) on a Sphere

We need to select a function that can assign a similarity measure to points in the sample space. This function for the regression setting is most naturally the Green's function of the Laplace-Beltrami operator in the geometry of the sample space (in this case the *unit sphere*).

Solve: $\left(\Delta_g - \frac{\partial}{\partial\tau}\right) G_\tau(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x}-\mathbf{x}')$, where $\Delta_g = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$ $\quad \mathbf{x} = (\theta,\phi)$, $\mathbf{x}' = (\theta',\phi')$

Natural kernel → Diffusion Green's function, *G*, on sphere → Covariance function, *K*, for GPR

$$G_\tau\left(\theta,\phi \mid \theta'=0,\phi'\right) = \sum_{l=0}^{\infty}\sqrt{\tfrac{2l+1}{4\pi}}e^{-l(l+1)\tau}\cdot Y_{l0}(\theta,\phi)$$

$$G_\tau\left(\gamma\right) = \sum_{l=0}^{\infty}\sqrt{\tfrac{2l+1}{4\pi}}e^{-l(l+1)\tau}\cdot\sqrt{\tfrac{2l+1}{4\pi}}P_l(\cos\gamma)$$

$$G_\tau\left(\theta,\phi \mid \theta',\phi'\right) = \sum_{l=0}^{\infty}e^{-l(l+1)\tau}\cdot\sum_{m=-l}^{l}Y_{lm}(\theta',\phi')\cdot Y_{lm}(\theta,\phi)$$

$$\lim_{\tau\to 0} G_\tau\left(\theta,\phi \mid \theta',\phi'\right) = \delta(\theta-\theta')\cdot\delta(\phi-\phi')$$

$$Y_{lm}(\theta,\phi) = \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}\cdot P_{lm}(\cos\theta)e^{im\phi}$$

$$Y_{l0}(\theta,\phi) = \sqrt{\frac{(2l+1)}{4\pi}}\cdot P_l(\cos\theta)$$

$$\cos\gamma = \cos(\phi-\phi')\cdot\sin\theta\sin\theta' + \cos\theta\cos\theta'$$

$$P_l(\cos\gamma) = \frac{4\pi}{2l+1}\sum_{m=-l}^{l}Y_{lm}(\theta',\phi')\cdot Y_{lm}(\theta,\phi)$$

$$\delta(\theta-\theta')\cdot\delta(\phi-\phi') = \sum_{l=0}^{\infty}\frac{2l+1}{4\pi}\cdot P_l(\cos\gamma)$$

$$= \sum_{l=0}^{\infty}\sum_{m=-l}^{l}Y_{lm}(\theta',\phi')\cdot Y_{lm}(\theta,\phi)$$

"Dot-product" covariance function (rotation invariant):

$$\boxed{G_\tau\left(X(\theta,\phi) \mid X(\theta',\phi')\right) = \sum_{l=0}^{\infty}e^{-l(l+1)\tau}\cdot\sum_{m=-l}^{l}Y_{lm}(\theta,\phi)\cdot Y_{lm}(\theta',\phi')}$$

Remark: To manage the inhomogeneous sampling in the data space, it may help to vary the kernel bandwidth, τ, so that each prediction depends on the *same* number of weighted samples. But if we let τ be position dependent this the kernel appears to no longer be symmetric. To restore this symmetry consider a "detailed-balance" approach to find self-consistent sample weights, w_i.

Gaussian Process Regression (GPR) on a Sphere

Green's function:

This accelerates the computation.

$$G_\tau\left(\theta_i,\phi_i \mid \theta_j,\phi_j\right)=\frac{1}{4\pi}\sum_{l=0}^{\infty}(2l+1)\cdot e^{-l(l+1)\tau}\cdot P_l(\cos\gamma_{ij})$$

$$\cos\gamma_{ij}=\cos(\phi_i-\phi_j)\cdot\sin\theta_i\sin\theta_j+\cos\theta_i\cos\theta_j$$

Recursion formula for Legendre polynomials:

$$P_l(x)=\frac{2l-1}{l}x\cdot P_{l-1}(x)-\frac{l-1}{l}P_{l-2}(x)$$

If $i=j$ then $$G_\tau\left(\theta_i,\phi_i \mid \theta_i,\phi_i\right)=\frac{1}{4\pi}\sum_{l=0}^{\infty}(2l+1)\cdot e^{-l(l+1)\tau}$$

NOTE: This is used to compute $K(X_*,X_*)$.

Covariance function:

noise variance of the ith sample (can be zero w/o loss of numerical stability)

$$\overset{(N+1)\times(N+1)}{K(\mathbf{X},\mathbf{X}\mid\tau,\sigma^2)}=\begin{bmatrix}\overset{N\times N}{G_\tau\left(\theta_i,\phi_i \mid \theta_j,\phi_j\right)+\sigma_i^2\delta_{ij}} & \overset{N\times 1}{G_\tau\left(\theta_i,\phi_i \mid \theta_*,\phi_*\right)}\\ \underset{1\times N}{G_\tau\left(\theta_*,\phi_* \mid \theta_j,\phi_j\right)} & \underset{1\times 1}{G_\tau\left(\theta_*,\phi_* \mid \theta_*,\phi_*\right)}\end{bmatrix}$$

Gaussian process that generates both training, Y, and testing data, Y_*:

$$\begin{bmatrix}Y\\ Y_*\end{bmatrix}\sim\mathcal{N}\left(\mathbf{0},\begin{bmatrix}G_\tau\left(\theta_i,\phi_i \mid \theta_j,\phi_j\right) & G_\tau\left(\theta_i,\phi_i \mid \theta_*,\phi_*\right)\\ G_\tau\left(\theta_*,\phi_* \mid \theta_j,\phi_j\right) & G_\tau\left(\theta_*,\phi_* \mid \theta_*,\phi_*\right)\end{bmatrix}\right)$$

GPR equations:

$$p\left(Y_* \mid X_*;X,Y\right)=\mathcal{N}\left(Y_* \mid \text{mean}(Y_*),\text{cov}(Y_*)\right)$$

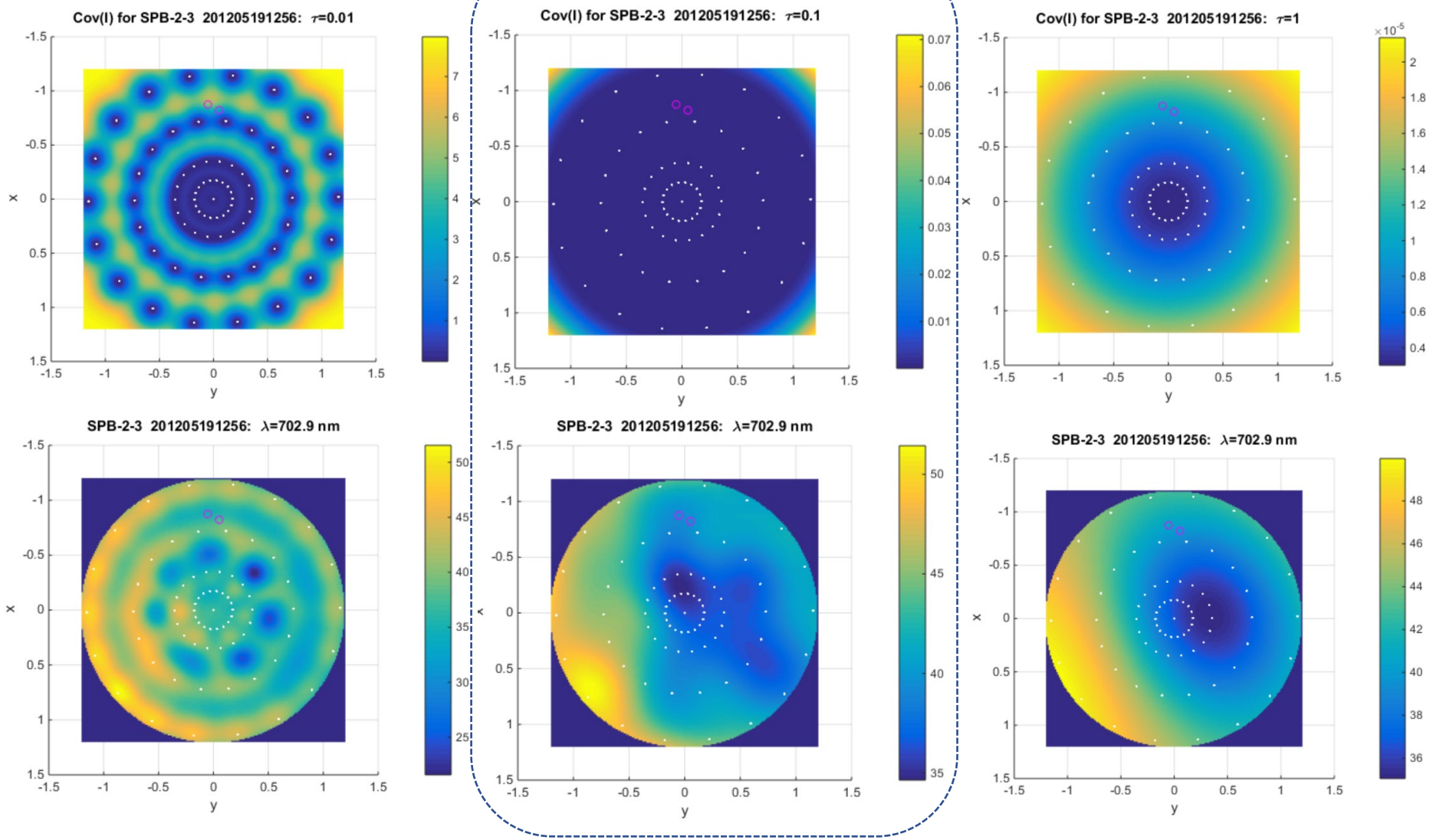
$$\text{mean}(Y_*)=K(X_*,X)\cdot\left[K(X,X)+\sigma^2(X)\cdot\mathbf{I}_n\right]^{-1}\cdot Y$$

$$\text{cov}(Y_*)=K(X_*,X_*)-K(X_*,X)\cdot\left[K(X,X)+\sigma_X^2\mathbf{I}_n\right]^{-1}\cdot K(X,X_*)$$

Gaussian Process Regression (GPR) on a Sphere

Good choice: τ=0.1

$\sigma \sim 0.1$ in sampling region



Expand GPR mean function on Y_{lm}'s

Observation points: $x_i = (\theta_i, \phi_i),\ i = 1, \ldots, N$

$$y_i = f(x_i)$$

Interpolation points: $x_* = (\theta, \phi)$

Estimate the function: $f(\mathbf{x}_*)$

Spherical Harmonic Kernel (Legendre form):

$$K_\tau(\theta, \phi \mid \theta', \phi') = \frac{1}{4\pi} \sum_{l=0}^{\infty} (2l+1) \cdot e^{-l(l+1)\tau} \cdot P_l(\cos\gamma)$$

$$\cos\gamma = \cos(\phi - \phi') \cdot \sin\theta \sin\theta' + \cos\theta \cos\theta'$$

Use this identity to convert to Y_{lm} form of kernel:

$$P_l(\cos\gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}(\theta', \phi') \cdot Y_{lm}(\theta, \phi)$$

$$K_\tau(\theta, \phi \mid \theta', \phi') = \sum_{l=0}^{\infty} e^{-l(l+1)\tau} \cdot \sum_{m=-l}^{l} Y_{lm}(\theta, \phi) \cdot Y_{lm}(\theta', \phi')$$

$$[\mathbf{k}(x_*)]_i = \sum_{l=0}^{\infty} e^{-l(l+1)\tau} \sum_{m=-l}^{l} Y_{lm}(\theta, \phi) \cdot Y_{lm}(\theta_i, \phi_i)$$

GPR mean function formula:

$$\bar{f}(\mathbf{x}_*) = \mathbf{k}_*^T (\mathbf{K}_\tau + \sigma_n^2 \mathbf{I}_N)^{-1} \mathbf{y}$$

$$= \sum_{i=1}^{N} [\mathbf{k}(x_*)]_i \cdot \left[(\mathbf{K}_\tau + \sigma_n^2 \mathbf{I}_N)^{-1} \mathbf{y} \right]_i$$

$$\bar{f}_L(\theta, \phi) = \sum_{i=1}^{N} \sum_{l=0}^{L} e^{-l(l+1)\tau} \sum_{m=-l}^{l} Y_{lm}(\theta, \phi) \cdot Y_{lm}(\theta_i, \phi_i) \left[(\mathbf{K}_\tau + \sigma_n^2 \mathbf{I}_N)^{-1} \mathbf{y} \right]_i$$

$$\bar{f}_L(\theta, \phi) = \sum_{l=0}^{L} \sum_{m=-l}^{l} a_{lm}(\tau) \cdot Y_{lm}(\theta, \phi)$$

$$a_{lm}(\tau) = e^{-l(l+1)\tau} \cdot \sum_{i=1}^{N} Y_{lm}(\theta_i, \phi_i) \cdot \left[(\mathbf{K}_\tau + \sigma_n^2 \mathbf{I}_N)^{-1} \mathbf{y} \right]_i$$

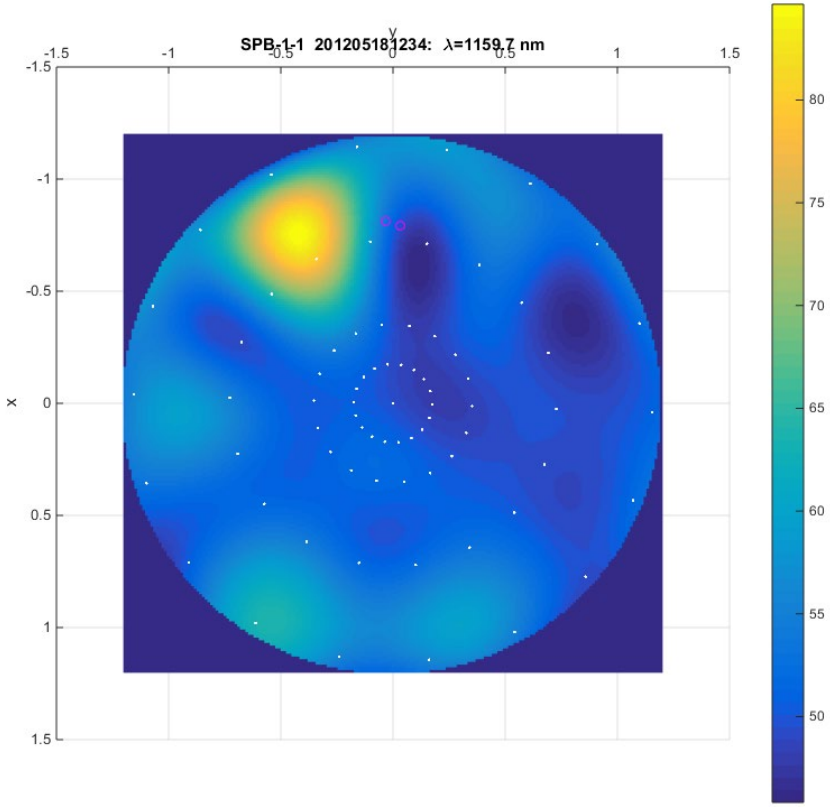
To find the "best" cut-off, L^*, of the spherical harmonic expansion, perform a minimax against the $\sigma(\tau)$:

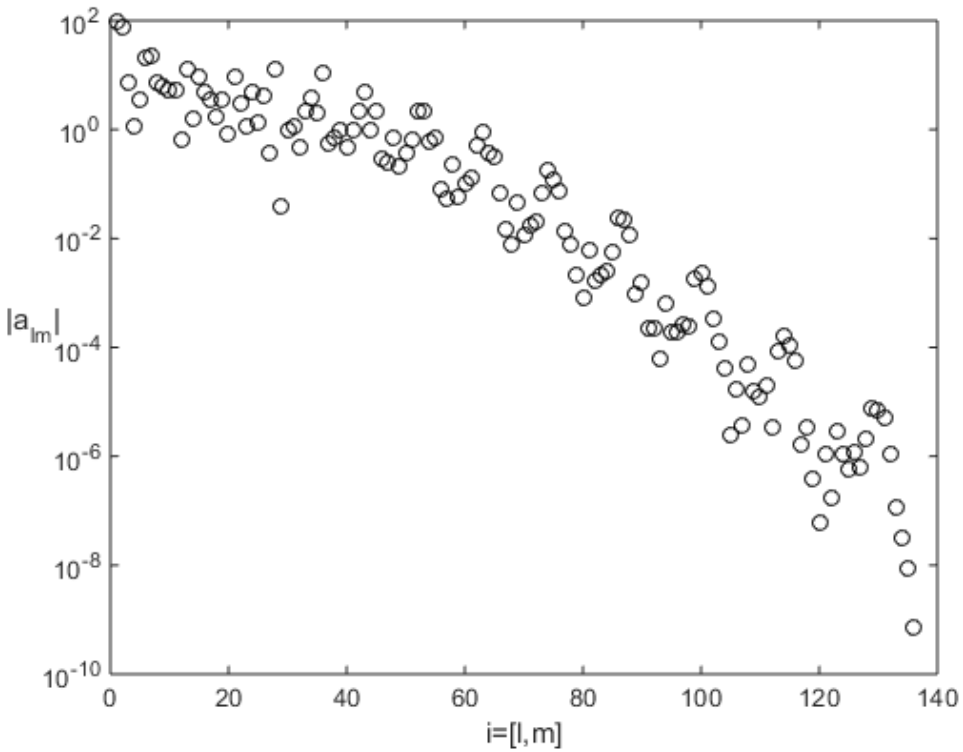
$$L^*(\tau) = \inf_L \left[\left(\sup_{\mathbf{x}_*} \left| \bar{f}_L(\theta, \phi) - \bar{f}(\mathbf{x}_*) \right| \right) < \sigma(\tau) \right]$$

GPR for Spherical Harmonics

$$\overline{f}_L(\theta,\phi) = \sum_{l=0}^{L} \sum_{m=-l}^{l} a_{lm}(\tau) \cdot Y_{lm}(\theta,\phi)$$

$$a_{lm}(\tau) = e^{-l(l+1)\tau} \cdot \sum_{i=1}^{N} Y_{lm}(\theta_i,\phi_i) \cdot \left[(\mathbf{K}_\tau + \sigma_n^2 \mathbf{I}_N)^{-1} \mathbf{y} \right]_i$$





Information Theory Review

Learning by diffusing: The information potential

Physics problem:

N particles are diffusing within an unknown potential energy function, $u(\boldsymbol{x})$.

→ Estimate $u(\boldsymbol{x})$ from a single observation of the N particle positions

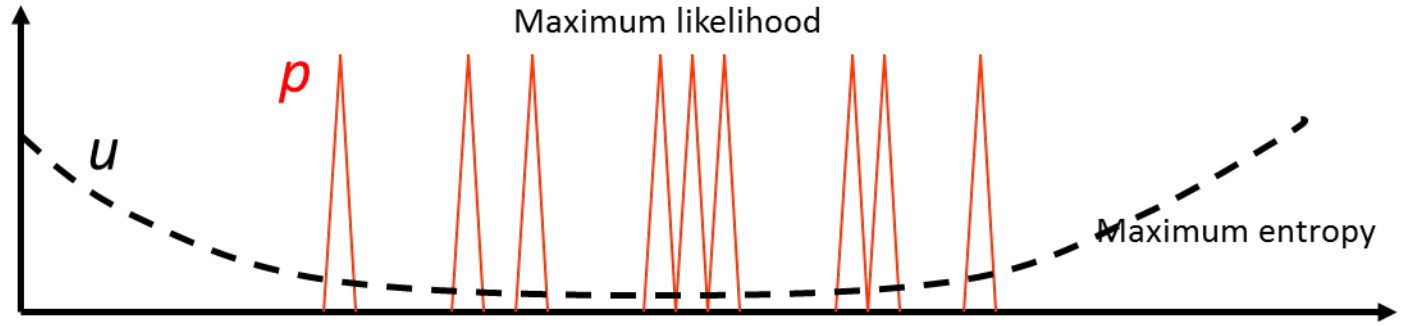
Drift-diffusion Eqn.:

$$\partial_t p = \nabla \cdot \left(D \nabla p + p \nabla u \right)$$

Empirical distribution as IC:

$$p(\mathbf{x},0) = \frac{1}{N} \sum_{n=1}^{N} \delta(\mathbf{x} - \mathbf{x}_n)$$

$$\lim_{t \to \infty} p(\mathbf{x},t) = \frac{e^{-u(\mathbf{x})/D}}{Z} \quad \Rightarrow \quad u(\mathbf{x}) = -\ln p(\mathbf{x}) + \ln Z$$



Probability and Information

Non-parametric approach Data: $p(\vec{x})$ Model: $q(\vec{x})$

Self-information

$$i_p(\vec{x}) = \log_2 \frac{1}{p(\vec{x})}$$

Average over $p(\vec{x})$

Entropy

$$H = \left\langle i_p(\vec{x}) \right\rangle_p$$
$$= -\int d\vec{x}\, p(\vec{x}) \log_2 p(\vec{x})$$

Relative Self-information

$$\delta_{q,p}(\vec{x}) = i_q(\vec{x}) - i_p(\vec{x})$$
$$= \log_2 \frac{p(\vec{x})}{q(\vec{x})}$$

Average over $p(\vec{x})$

Relative entropy

$$D_{KL}(p \,||\, q) = \left\langle \delta_{q,p}(\vec{x}) \right\rangle_p$$
$$= \int d\vec{x}\, p(\vec{x}) \cdot \log_2 \frac{p(\vec{x})}{q(\vec{x})}$$

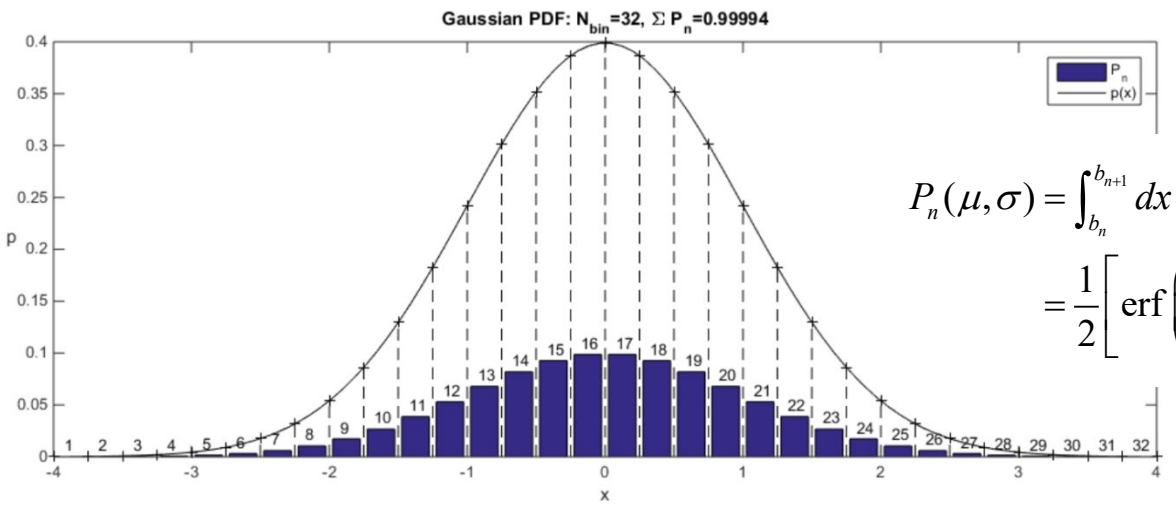
Provides a way to *compare* probability models.

a.k.a, Kullback-Leibler divergence

Probability and Information: Gaussian PDF

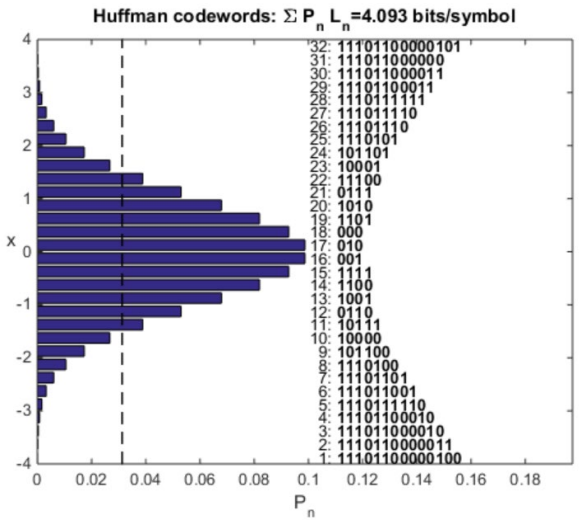
Probability density function: $p(x \mid \mu,\sigma) = (2\pi\sigma^2)^{-1/2} e^{-(x-\mu)^2/2\sigma^2}$

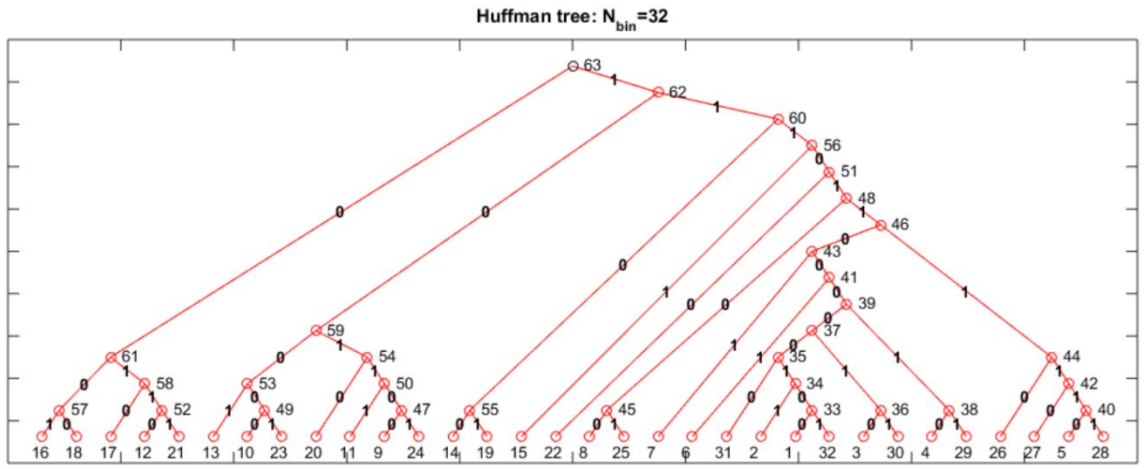
Codeword length (bits): $i_p(n) = \log_2 \dfrac{1}{P_n(\mu,\sigma)}$



$$P_n(\mu,\sigma) = \int_{b_n}^{b_{n+1}} dx\, p(x \mid \mu,\sigma)$$

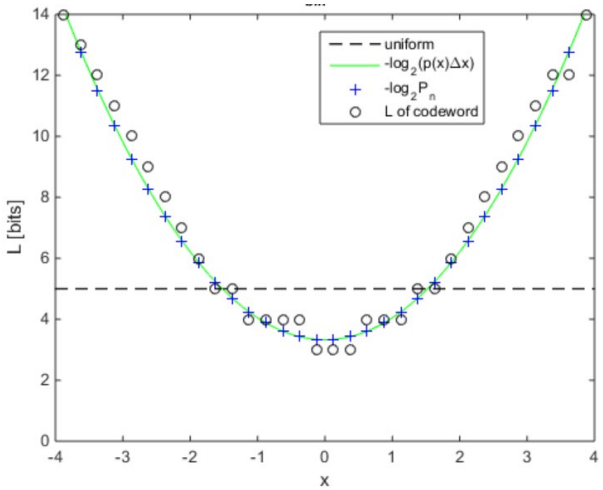
$$= \frac{1}{2}\left[\operatorname{erf}\left(\frac{b_{n+1}-\mu}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{b_n-\mu}{\sqrt{2}\sigma}\right)\right]$$





$$P_n = 2^{-L_n}$$

$$\sum_{n=1}^{N} 2^{-L_n} \le 1$$



Entropy of Dependent Random Variables

The joint entropy of X, Y is:

$$H(X,Y) = \sum_{xy \in \mathcal{A}_X \mathcal{A}_Y} P(x,y) \log \frac{1}{P(x,y)}. \tag{8.1}$$

Entropy is additive for independent random variables:

$$H(X,Y) = H(X) + H(Y) \text{ iff } P(x,y) = P(x)P(y). \tag{8.2}$$

The conditional entropy of X given $y = b_k$ is the entropy of the probability distribution $P(x \mid y = b_k)$.

$$H(X \mid y = b_k) \equiv \sum_{x \in \mathcal{A}_X} P(x \mid y = b_k) \log \frac{1}{P(x \mid y = b_k)}. \tag{8.3}$$

The conditional entropy of X given Y is the average, over y, of the conditional entropy of X given y.

$$\begin{aligned} H(X \mid Y) &\equiv \sum_{y \in \mathcal{A}_Y} P(y) \left[\sum_{x \in \mathcal{A}_X} P(x \mid y) \log \frac{1}{P(x \mid y)} \right] \\ &= \sum_{xy \in \mathcal{A}_X \mathcal{A}_Y} P(x,y) \log \frac{1}{P(x \mid y)}. \end{aligned} \tag{8.4}$$

This measures the average uncertainty that remains about x when y is known.

The marginal entropy of X is another name for the entropy of X, $H(X)$, used to contrast it with the conditional entropies listed above.

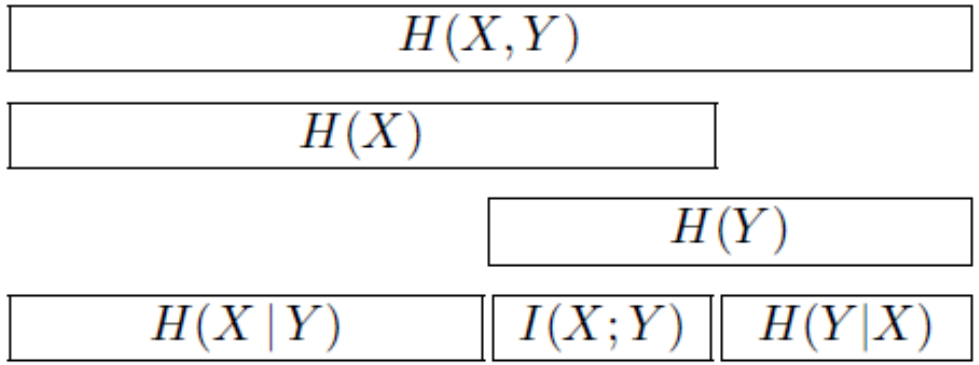
Chain Rule for Entropy: $H(X,Y) = H(X) + H(Y \mid X)$

$$= H(Y) + H(X \mid Y)$$

Proof,

$$\begin{aligned} H(X,Y) &= -\sum_{x,y} p(x,y) \cdot \log p(x,y) \\ &= -\sum_{x,y} p(x,y) \cdot \log p(y \mid x) p(x) \\ &= -\sum_{x,y} p(x,y) \cdot \log p(x) - \sum_{x,y} p(x,y) \cdot \log p(y \mid x) \\ &= H(X) + H(Y \mid X) \end{aligned}$$

Useful diagram to remember entropy relations:



Quantify probabilistic dependence: Mutual Information

How can we measure the information shared between two random variables X and Y?

Answer: The **extra** information is in the joint probability density function (PDF), $p(x,y)$, beyond what is in the product of the marginal distributions, $p(x) \cdot p(y)$.

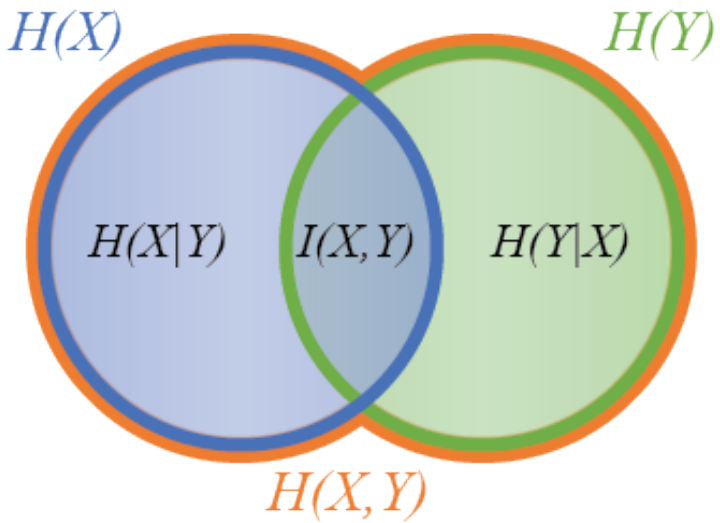
Mutual information: $I(X,Y) = D_{KL}(p(x,y) \,||\, p(x) \cdot p(y))$

$$I(X,Y) = \sum_{y \in Y} \sum_{x \in X} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \geq 0$$

if X and Y are independent then $p(x,y) = p(x)p(y)$ and $I(X,Y) = 0$.

Mutual information

$$\begin{aligned} I(X,Y) &= H(X) - H(X \mid Y) \\ &= H(Y) - H(Y \mid X) \\ &= H(X) + H(Y) - H(X,Y) \end{aligned}$$

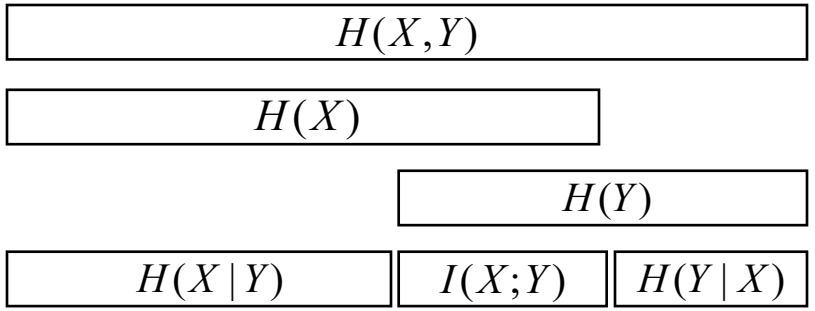


Maximum Possible Mutual Information

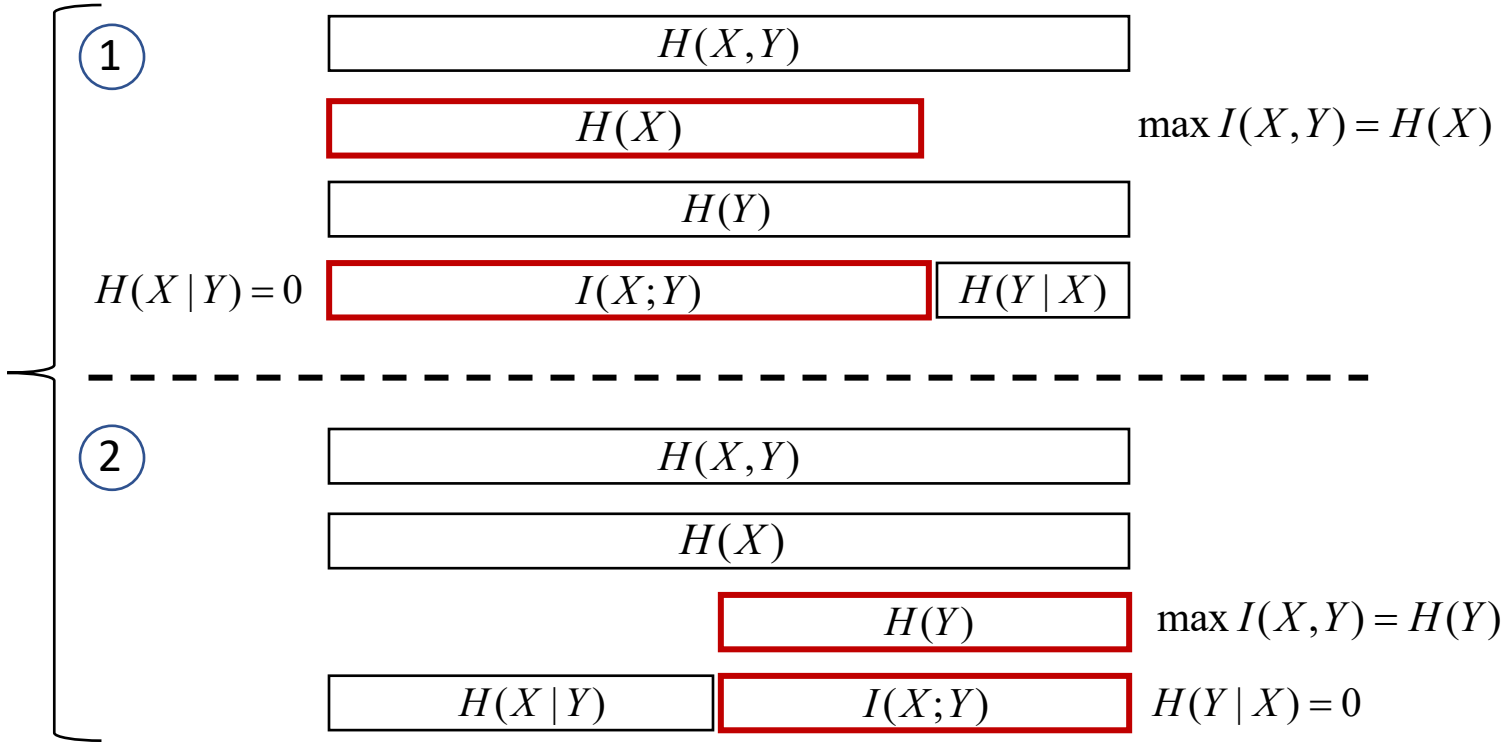
Upper bound on the mutual information, $I(X;Y)$, is found by either setting $H(X|Y)=0$ *or* $H(Y|X)=0$.

$$\max I(X,Y) = \min\{H(X), H(Y)\}$$

$$I(X,Y) = H(X) - H(X \mid Y)$$



$$I(X,Y) = H(Y) - H(Y \mid X)$$



Correlation: From Covariance to Mutual Information

$$\rho_{X,Y} = \operatorname{corr}(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\mathrm{E}[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}, \quad \text{if } \sigma_X \sigma_Y > 0.$$

$$X, Y \text{ independent} \quad \Rightarrow \quad \rho_{X,Y} = 0 \quad (X, Y \text{ uncorrelated})$$

$$\rho_{X,Y} = 0 \quad (X, Y \text{ uncorrelated}) \quad \not\Rightarrow \quad X, Y \text{ independent}$$

$$\mu_x = \sum_x \sum_y P(x,y) \cdot x \qquad \mu_y = \sum_x \sum_y P(x,y) \cdot y$$

$$\operatorname{cov}(X,Y) = \sum_x \sum_y P(x,y) \cdot (x - \mu_x)(y - \mu_y)$$

Rather contrived example of how *uncorrelated* does NOT imply probabilistic independence

Consider the joint probability distribution of X and Y given in the table below.

$\mathrm{P}(X = x, Y = y)$

x \ y	−1	0	1
0	0	$\frac{1}{3}$	0
1	$\frac{1}{3}$	0	$\frac{1}{3}$

For this joint distribution, the marginal distributions are:

$$\mathrm{P}(X = x) = \begin{cases} \frac{1}{3} & \text{for } x = 0 \\ \frac{2}{3} & \text{for } x = 1 \end{cases}$$

$$\mathrm{P}(Y = y) = \begin{cases} \frac{1}{3} & \text{for } y = -1 \\ \frac{1}{3} & \text{for } y = 0 \\ \frac{1}{3} & \text{for } y = 1 \end{cases}$$

This yields the following expectations and variances:

$$\mu_X = \frac{2}{3}$$
$$\mu_Y = 0$$
$$\sigma_X^2 = \frac{2}{9}$$
$$\sigma_Y^2 = \frac{2}{3}$$

Therefore:

$$\begin{aligned} \rho_{X,Y} &= \frac{1}{\sigma_X \sigma_Y} \mathrm{E}[(X-\mu_X)(Y-\mu_Y)] \\ &= \frac{1}{\sigma_X \sigma_Y} \sum_{x,y} (x-\mu_X)(y-\mu_Y) \mathrm{P}(X=x, Y=y) \\ &= \left(1 - \frac{2}{3}\right)(-1-0)\frac{1}{3} + \left(0 - \frac{2}{3}\right)(0-0)\frac{1}{3} + \left(1 - \frac{2}{3}\right)(1-0)\frac{1}{3} = 0. \end{aligned}$$

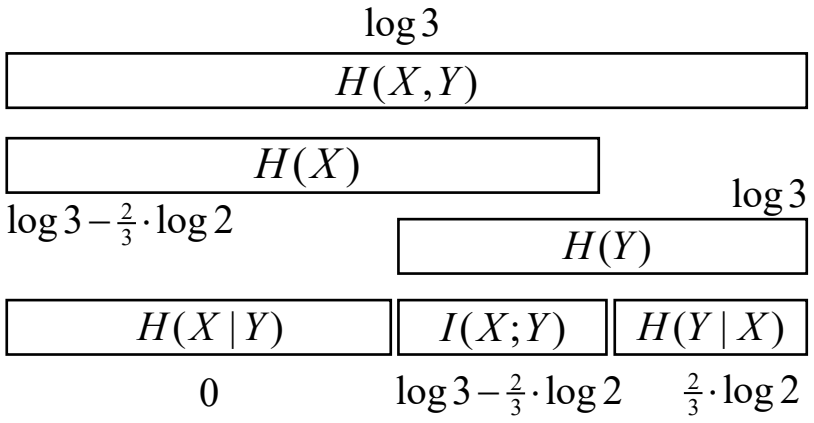
Check your understanding by computing the entropies below!

Correlation: From Covariance to Mutual Information

Now, we can see how mutual information detects the dependence.

$$\rho_{X,Y} = \frac{I(X,Y)}{\min\{H(X),H(Y)\}} = 1 - \frac{H(Y\,|\,X)}{H(Y)} = 1 - \frac{H(X\,|\,Y)}{H(X)}$$

$$\rho_{X,Y} = 0 \Leftrightarrow X, Y \text{ independent}$$



Consider the joint probability distribution of X and Y given in the table below.

$\mathrm{P}(X = x, Y = y)$

x \ y	−1	0	1
0	0	$\frac{1}{3}$	0
1	$\frac{1}{3}$	0	$\frac{1}{3}$

$$\rho_{X,Y} = 1$$

For this joint distribution, the marginal distributions are:

$$\mathrm{P}(X = x) = \begin{cases} \frac{1}{3} & \text{for } x = 0 \\ \frac{2}{3} & \text{for } x = 1 \end{cases}$$

$$\mathrm{P}(Y = y) = \begin{cases} \frac{1}{3} & \text{for } y = -1 \\ \frac{1}{3} & \text{for } y = 0 \\ \frac{1}{3} & \text{for } y = 1 \end{cases}$$

$$H(X,Y) = -\sum_{x,y} p(x,y)\cdot\log p(x,y)$$
$$= -0\cdot\log 0 - \tfrac{1}{3}\cdot\log\tfrac{1}{3} - 0\cdot\log 0$$
$$-\tfrac{1}{3}\cdot\log\tfrac{1}{3} - 0\cdot\log 0 - \tfrac{1}{3}\cdot\log\tfrac{1}{3} = \underline{\log 3}$$

$$H(X) = -\sum_{x} p(x)\cdot\log p(x) = -\tfrac{1}{3}\cdot\log\tfrac{1}{3} - \tfrac{2}{3}\cdot\log\tfrac{2}{3}$$
$$= \tfrac{1}{3}\cdot\log 3 + \tfrac{1}{3}\cdot\log\tfrac{9}{4} = \underline{\log 3 - \tfrac{2}{3}\cdot\log 2}$$

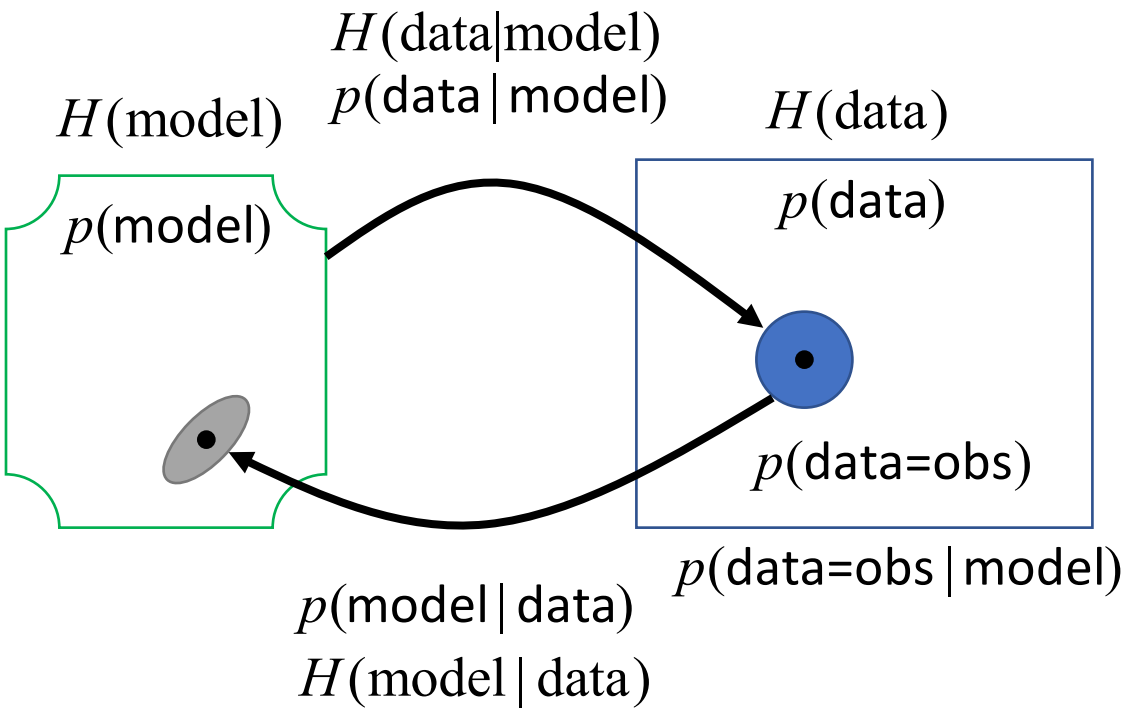
$$H(Y) = -\sum_{y} p(y)\cdot\log p(y)$$
$$= -\tfrac{1}{3}\cdot\log\tfrac{1}{3} - \tfrac{1}{3}\cdot\log\tfrac{1}{3} - \tfrac{1}{3}\cdot\log\tfrac{1}{3} = \underline{\log 3}$$

$$H(X\,|\,Y) = -\sum_{x,y} p(x,y)\cdot\log\frac{p(x,y)}{p(y)}$$
$$= -0\cdot\log\tfrac{0}{1/3} - \tfrac{1}{3}\cdot\log\tfrac{1/3}{1/3} - 0\cdot\log\tfrac{0}{1/3}$$
$$-\tfrac{1}{3}\cdot\log\tfrac{1/3}{1/3} - 0\cdot\log\tfrac{0}{1/3} - \tfrac{1}{3}\cdot\log\tfrac{1/3}{1/3} = \underline{0}$$

$$H(Y\,|\,X) = -\sum_{x,y} p(x,y)\cdot\log\frac{p(x,y)}{p(x)}$$
$$= -0\cdot\log\tfrac{0}{1/3} - \tfrac{1}{3}\cdot\log\tfrac{1/3}{1/3} - 0\cdot\log\tfrac{0}{1/3}$$
$$-\tfrac{1}{3}\cdot\log\tfrac{1/3}{2/3} - 0\cdot\log\tfrac{0}{2/3} - \tfrac{1}{3}\cdot\log\tfrac{1/3}{2/3}$$
$$= -\tfrac{2}{3}\cdot\log\tfrac{1/3}{2/3} = \underline{\tfrac{2}{3}\cdot\log 2}$$

Bayesian Inference and Information Theory

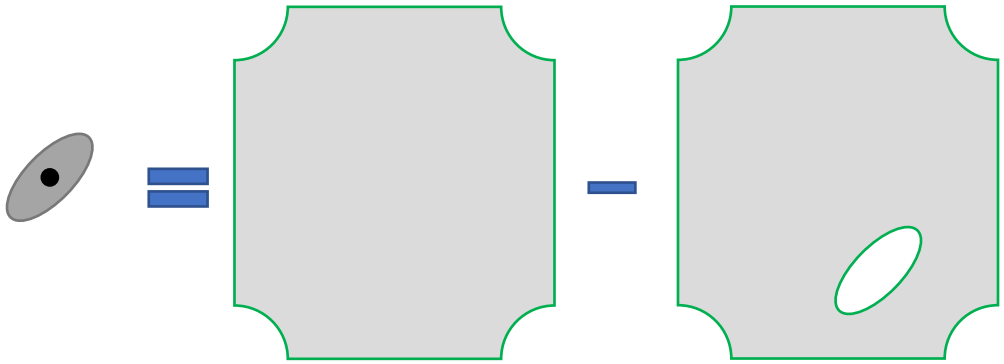
Conventional Bayes' rule:



NOTE: DATA space refers to the space of all *possible* observations.

Information-theoretic Bayes' rule:

$$H(\text{model}\,|\,\text{data}) = H(\text{model}) - I(\text{model},\text{data})$$



Optimization problem for Experimental Design:

$$I(\text{model},\text{data}) = H(\text{model}) - H(\text{model}\,|\,\text{data})$$

Maximize by selecting

$p(\text{data}\,|\,\text{model})$

Information Theory and Inference

Kullback-Leibler divergence: $D_{KL}(p \| q) = \left\langle u_q - u_p \right\rangle_{p(x)} = \sum_{x \in X} p(x) \log \frac{p(x)}{q(x)}$

Bayes' rule for inference: $p(x \mid y) = \frac{p(y \mid x) p(x)}{p(y)}$

Information gain: $D_{KL}\left(p(x \mid y) \| p(x)\right) = \sum_{x \in X} p(x \mid y) \log \frac{p(x \mid y)}{p(x)} = \frac{1}{p(y)} \sum_{x \in X} p(y \mid x) p(x) \log \frac{p(y \mid x)}{p(y)}$

$$= \frac{1}{p(y)} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x) p(y)}$$

Interpretation: The information gained about the distribution of X after observing Y=y.

Average relative entropy gain: $I(X, Y) = \sum_{y \in Y} p(y) D_{KL}\left(p(x \mid y) \| p(x)\right)$
(or *mutual information*)

Interpretation: The average information gained about the distribution of X by observing the distribution Y.

Information-theoretic Bayes' rule: $H(X \mid Y) = H(X) - I(X, Y)$
(w.r.t. average over all obs.)

Interpretation: Entropy remaining in X after observing the distribution of Y.

Optimization problem for Experimental Design: $\hat{p}(y \mid x) = \arg\min_{p(y|x)} H(X \mid Y) = \arg\max_{p(y|x)} I(X, Y)$

Information Theory and Inference

Kullback-Leibler divergence: $D_{KL}(p \| q) = \langle u_q - u_p \rangle_{p(x)} = \sum_{x \in X} p(x) \log \frac{p(x)}{q(x)}$

Bayes' rule for inference: $p(x \mid y) = \frac{p(y \mid x) p(x)}{p(y)}$

Let's use this!

Information gain: $D_{KL}\left(p(x \mid y) \| p(x)\right) = \sum_{x \in X} p(x \mid y) \log \frac{p(x \mid y)}{p(x)} = \frac{1}{p(y)} \sum_{x \in X} p(y \mid x) p(x) \log \frac{p(y \mid x)}{p(y)}$

$$= \frac{1}{p(y)} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x) p(y)}$$

Interpretation: The information gained about the distribution of X after observing Y=y.

Average relative entropy gain: $I(X, Y) = \sum_{y \in Y} p(y) D_{KL}\left(p(x \mid y) \| p(x)\right)$
(or *mutual information*)

Interpretation: The average information gained about the distribution of X by observing the distribution Y.

Information-theoretic Bayes' rule: $H(X \mid Y) = H(X) - I(X, Y)$
(w.r.t. average over all obs.)

Interpretation: Entropy remaining in X after observing the distribution of Y.

Optimization problem for Experimental Design: $\hat{p}(y \mid x) = \arg\min_{p(y|x)} H(X \mid Y) = \arg\max_{p(y|x)} I(X, Y)$

Using information theory to make a decision …

For a binary decision, how close are we to learning one bit?

$$p(\mathcal{M}_\mathrm{A}) = 0.5$$

$$p(\mathcal{M}_\mathrm{B}) = 0.5$$

$$p(\mathcal{M}_\mathrm{A} \mid \mathrm{data}) = \frac{\int_{\Omega_i} d\boldsymbol{\theta}\; p(\mathrm{data} \mid \boldsymbol{\theta}, \mathcal{M}_\mathrm{A}) \cdot p(\boldsymbol{\theta} \mid \mathcal{M}_\mathrm{A}) \cdot p(\mathcal{M}_\mathrm{A})}{p(\mathrm{data})}$$

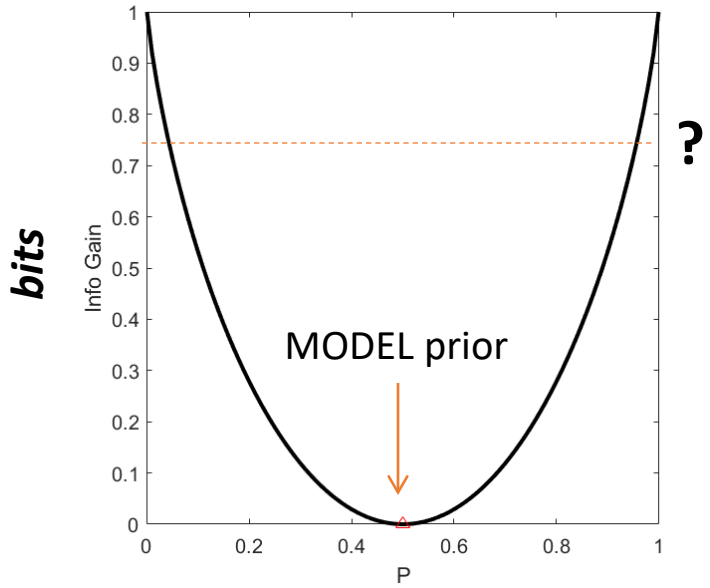
$$p(\mathcal{M}_\mathrm{B} \mid \mathrm{data}) = 1 - p(\mathcal{M}_\mathrm{A} \mid \mathrm{data})$$

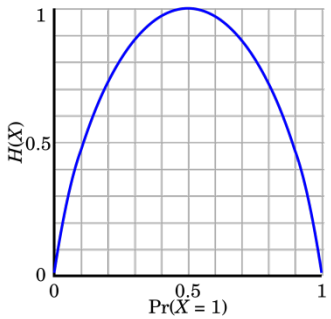
Information gain: $D_{KL}\left(p(\mathcal{M}_\mathrm{i} \mid \mathrm{data}) \,\|\, p(\mathcal{M}_\mathrm{i})\right) = \sum_{i=1,2} p(\mathcal{M}_\mathrm{i} \mid \mathrm{data}) \log_2 \frac{p(\mathcal{M}_\mathrm{i} \mid \mathrm{data})}{p(\mathcal{M}_\mathrm{i})}$

Interpretation: The information gained about the distribution of X after observing $Y=y$.

$$= P \cdot \log_2 \frac{P}{0.5} + (1-P) \cdot \log_2 \frac{1-P}{0.5}, \text{ where } P \doteq p(\mathcal{M}_\mathrm{A} \mid \mathrm{data})$$

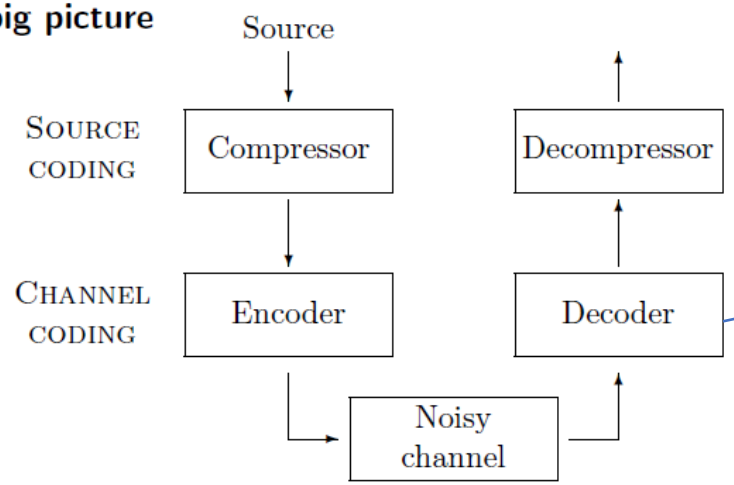
$$= 1 + P \cdot \log_2 P + (1-P) \cdot \log_2 (1-P) = 1 - H_{\mathrm{binary}}(P)$$





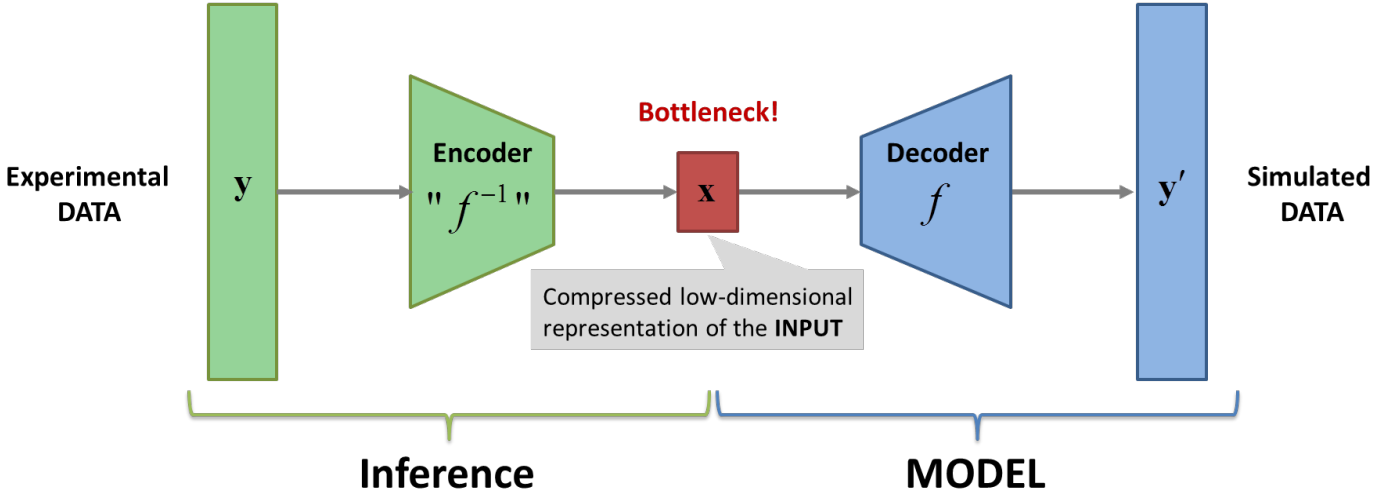
Communications Channel

The big picture



The optimal decoder for a channel code is the one that minimizes the probability of block error. It decodes an output $\mathbf{y}$ as the input s that has maximum posterior probability $P(s \mid \mathbf{y})$.

$$P(s \mid \mathbf{y}) = \frac{P(\mathbf{y} \mid s)P(s)}{\sum_{s'} P(\mathbf{y} \mid s')P(s')} \tag{9.17}$$

$$\hat{s}_{\text{optimal}} = \operatorname{argmax} P(s \mid \mathbf{y}). \tag{9.18}$$

A uniform prior distribution on s is usually assumed, in which case the optimal decoder is also the *maximum likelihood decoder*, i.e., the decoder that maps an output $\mathbf{y}$ to the input s that has maximum *likelihood* $P(\mathbf{y} \mid s)$.